Financial Engineering - Assignment RM 5

Group 6

Affine form of the discount factor

We start from the forward discount factor formula given in the OU framework:

$$\frac{B(t,T_1,T_2)}{B(t_0,T_1,T_2)} = \exp\left(-x_t\left[\sigma(t,T_2)-\sigma(t,T_1)\right] - \frac{1}{2}\int_{t_0}^{t}\left[\sigma^2(u,T_2)-\sigma^2(u,T_1)\right]du\right)$$

where $\sigma(u,\tau) = \dfrac{\sigma}{a}\left(1-e^{-a(\tau-u)}\right)$.

Set $T_1 = s$, $T_2 = \tau$, and evaluate at $t = s$:

$$\frac{B(s,s,\tau)}{B(t_0,s,\tau)} = \exp\left(-x_s\left[\sigma(s,\tau)-\sigma(s,s)\right] - \frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)$$

Since $\sigma(s,s) = \dfrac{\sigma}{a}\left(1-e^{-a(s-s)}\right) = 0$, and noting that $B(s,s,\tau) = B(s,\tau)$ (spot discount factor at time s):

$$\frac{B(s,\tau)}{B(t_0,s,\tau)} = \exp\left(-x_s\,\sigma(s,\tau) - \frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)$$

Using the definition of the forward discount factor $B(t_0,s,\tau) = \dfrac{B(t_0,\tau)}{B(t_0,s)}$:

$$B(s,\tau) = \frac{B(t_0,\tau)}{B(t_0,s)}\exp\left(-\frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)\cdot\exp\left(-x_s\,\sigma(s,\tau)\right)$$

Identifying $C(s,\tau) = \sigma(s,\tau) = \dfrac{1-e^{-a(\tau-s)}}{a}$ and:

$$A(s,\tau) = \frac{B(t_0,\tau)}{B(t_0,s)}\exp\left(-\frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)$$

we obtain the affine form:

$$\boxed{B(s,\tau) = A(s,\tau)\,e^{-x_s C(s,\tau)}}$$

Results and Discussion

Base Case: EPE and PFE

The base case corresponds to a vanilla 10-year payer IRS entered on February 15th, 2008, with no credit mitigation. The EPE and PFE profiles exhibit a characteristic sawtooth pattern throughout the life of the swap: exposure peaks just before each annual coupon payment date, when the present value of remaining fixed cashflows is highest, and drops sharply immediately after each payment, as one coupon is removed from the remaining cashflow schedule. This pattern is a direct consequence of the discrete coupon structure of the IRS and is economically expected. The sawtooth is more pronounced in the EPE than in the PFE, since the latter is driven by the tail of the distribution and is less sensitive to the average behaviour around coupon dates.

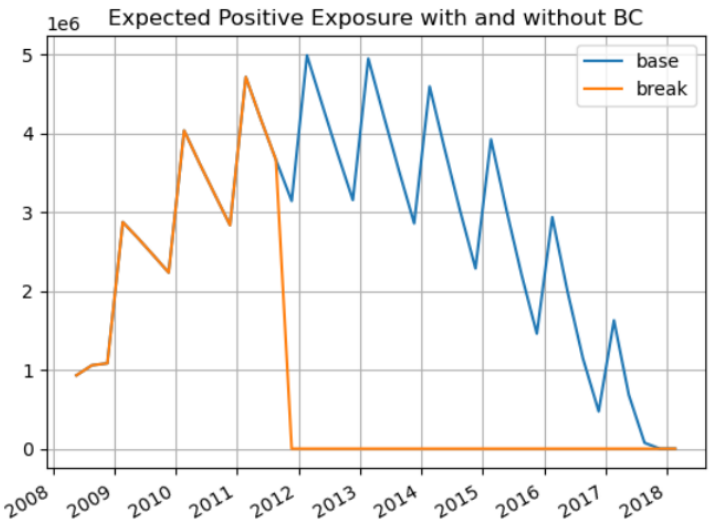
The smooth envelope is the result of two competing mechanisms acting on different time scales:

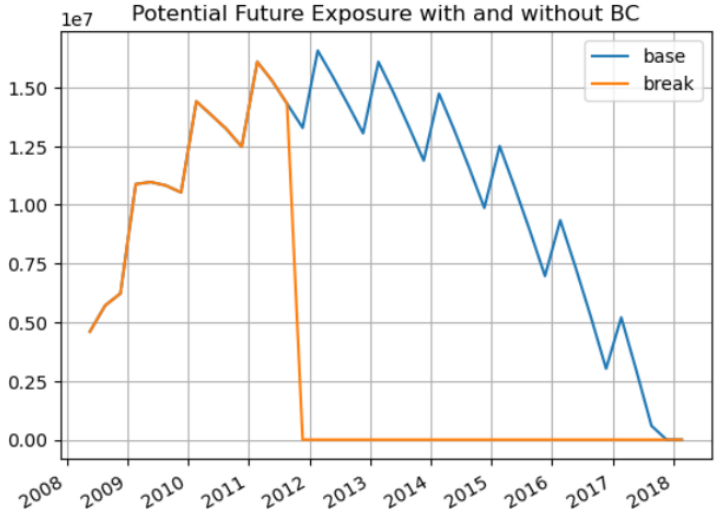
- **Diffusion effect.** The variance of the short rate process x_t grows as $\frac{\sigma^2}{2a}\left(1 - e^{-2at}\right)$, i.e. roughly with $\sqrt{t}$ in the early life of the swap. This widens the distribution of the future MtM and pushes both EPE and PFE upwards.
- **Amortisation effect.** As time elapses, the residual tenor of the swap shrinks. Fewer remaining cashflows mean that even large rate moves translate into smaller MtM impact, which pulls both EPE and PFE downwards.

The interplay of the two effects generates the typical hump shape, peaking around mid-life of the swap. The Peak-PFE at 95% reaches approximately 1.65×10^7 EUR.

Break Clause at 3.5 Years

The introduction of a mandatory break clause at 3.5 years (mid-2011) eliminates all exposure beyond that date. The EPE and PFE break curves are identical to the base case up to the break date, then collapse to zero immediately after, since the position is assumed terminated upon exercise. This confirms the effectiveness of break clauses as a duration-limiting risk mitigation tool. The Peak-PFE under the break clause scenario is significantly lower than the base case, bounded by the maximum exposure achievable within the 3.5-year window.





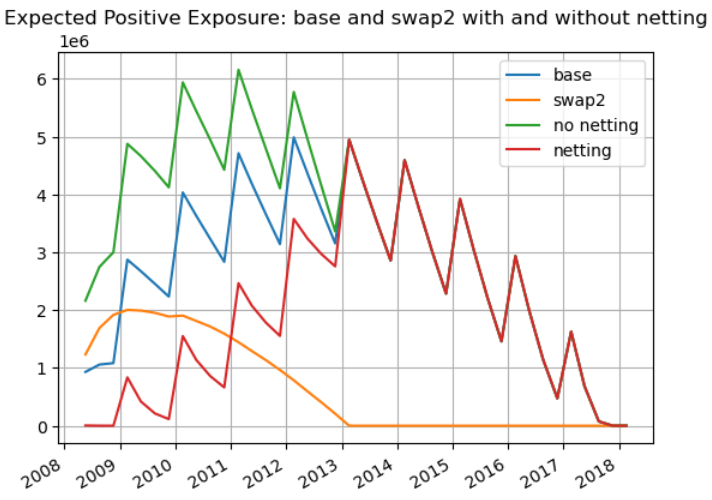
Netting Agreement

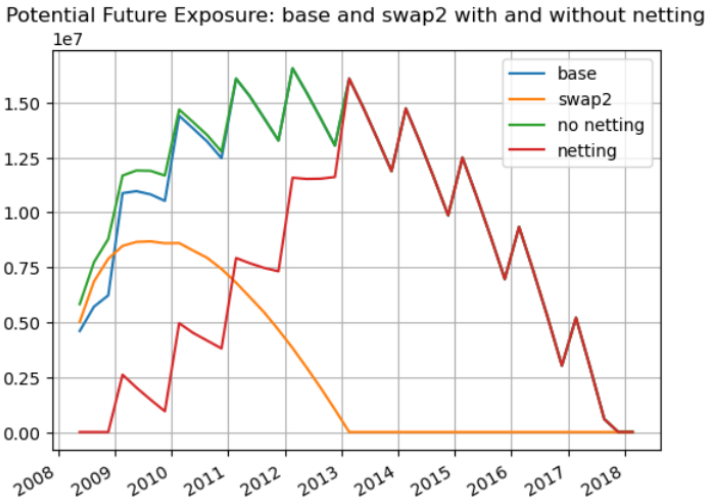
The netting analysis compares four quantities at each simulation date: the standalone exposure of swap 1 (base), the standalone exposure of swap 2, their sum (no netting), and the positive part of their combined MtF (netting). The theoretical inequality

$$\mathrm{EPE}_{\text{netting}}(s) \leq \mathrm{EPE}_{\text{no netting}}(s)$$

is verified, since $\mathbb{E}[\max(V_1 + V_2, 0)] \leq \mathbb{E}[\max(V_1, 0)] + \mathbb{E}[\max(V_2, 0)]$.

The netted EPE lies *above* the standalone base curve over the $[0, 5\text{y}]$ window. This is not due to a simultaneous in-the-money condition (the payer and the receiver are directionally opposite in rates by construction), but to the *notional asymmetry*: the receiver carries 150 M EUR against the 100 M EUR of the payer, so in rate-falling scenarios the netted MtM is dominated by the receiver and exceeds in magnitude what the payer alone produces in symmetric rate-rising scenarios. Sub-additivity with respect to the no-netting sum is nevertheless preserved. After swap 2 matures (around 2013), the netted and no-netting curves collapse back onto the base curve.





CSA Agreement with Annual Margin Calls

The collateral parameters are chosen asymmetrically in favor of Polimi Bank (party B):

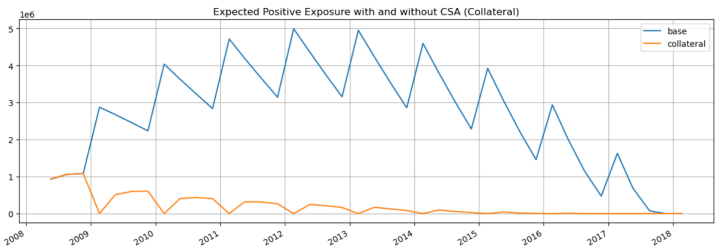
- $\mathrm{THR}_B \to +\infty$: by design, B is never required to post collateral;
- $\mathrm{THR}_C = 0$, $\mathrm{MTA}_C = 0$: the counterparty posts as soon as any positive exposure arises;
- $\mathrm{IA}_B = \mathrm{IA}_C = 0$: no independent amount.

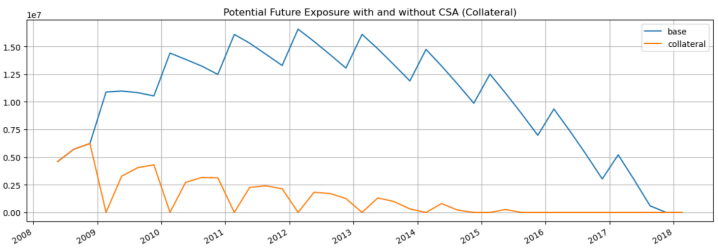
The bilateral CSA logic is implemented in `update_collateral`, which computes a scenario-wise target variation margin

$$\mathrm{VM}^* \;=\; \max(\mathrm{MtM} - \mathrm{THR}_C, 0) \cdot \mathbf{1}_{\mathrm{MtM}\geq 0} \;+\; \min(\mathrm{MtM} + \mathrm{THR}_B, 0) \cdot \mathbf{1}_{\mathrm{MtM}<0},$$

applies the MTA filter on $\Delta\mathrm{VM} = \mathrm{VM}^* - \mathrm{VM}$, and finally enforces the caps. With the parameters above this collapses to "the counterparty posts collateral equal to $\max(\mathrm{MtM}, 0)$ at every annual posting date", while Polimi Bank never posts. Between two posting dates the posted collateral is remunerated at the shortest-tenor zero rate available on the simulated HW curve, used as a proxy for the path-wise risk-free rate.

The resulting EPE with collateral is much lower at every simulation date, confirming the effectiveness of the CSA. The residual exposure visible in the plots arises from the *margin period of risk*: between two consecutive posting dates the MtM moves freely while the posted collateral is frozen (modulo risk-free remuneration), so the exposure can re-accumulate. The size of the residual exposure scales approximately as $\sigma\sqrt{\Delta t_{\mathrm{call}}}$ with the inter-call window Δt_{call}: with monthly, weekly or daily margining, the residual would shrink by a factor $\sqrt{12}$, $\sqrt{52}$, $\sqrt{252}$ respectively. The PFE with collateral remains well below 10^6 throughout, compared with a Peak-PFE of 1.65×10^7 in the base case, a reduction of more than 90%.





CVA Estimation

Under the assumptions of unilateral default risk, independence between exposure and default, flat hazard rate $\lambda = 200$ bps and LGD $= 60\%$, the CVA is computed as:

$$\text{CVA} = \text{LGD} \sum_{i=1}^{N} \text{EPE}(t_i) \cdot B(0, t_i) \cdot \left(e^{-\lambda t_{i-1}} - e^{-\lambda t_i} \right)$$

Scenario	CVA (EUR)	CVA (M EUR)
Base	251,722	0.252
Break clause	106,107	0.106
Swap 2 standalone	71,929	0.072
No netting	323,651	0.324
Netting	164,570	0.165
Collateral (CSA)	22,152	0.022

Table 1: CVA estimates across mitigation scenarios (main-loop engine).

The no-netting CVA (323,651) equals the sum of the two standalone CVAs (251,722 + 71,929), confirming CVA additivity without netting. The netting agreement reduces CVA by 49%, the break clause by 58%, and the CSA by 91%. The residual CVA under the CSA reflects the margin period of risk between annual margin calls.

CVA Sensitivity Analysis

Three first-order sensitivities are computed by bump-and-reprice (+1bp), targeting the main market risk factors of the CVA: counterparty credit risk (CR Delta), interest-rate risk (IR Delta) and model volatility risk (Vega). Base and bumped CVAs are computed with the *same* pricing function (`compute_epe_base`) and the same random numbers (common random numbers), so that the difference isolates the effect of the bump. Because `compute_epe_base` still contains a small residual coupon-double-count bias (see the discussion at the end of this section), its base CVA for the payer swap is 207,600 EUR instead of the 251,722 EUR obtained with the main Monte Carlo loop of Table 1; however the bias affects base and bumped computations identically and therefore cancels in the deltas shown below.

Sensitivity	Risk factor	CVA bumped (EUR)	ΔCVA / bp (EUR)
CR Delta	Hazard rate λ	208,552	+952
IR Delta	Parallel curve shift	209,273	+1,673
Vega	HW volatility σ	210,331	+2,731

Table 2: CVA sensitivities (base CVA for the sensitivity engine = EUR 207,600, bump = +1bp).

All three sensitivities are positive and of the same order of magnitude. The sign follows from the structure of the CVA: it is a positive-weighted sum of EPEs, $\text{CVA} = \text{LGD} \sum_i w_i \, \text{EPE}(t_i)$ with $w_i = B(0, t_i) \left(e^{-\lambda t_{i-1}} - e^{-\lambda t_i} \right) > 0$, and each $\text{EPE}(t) = \mathbb{E}[\max(\text{MtM}_t, 0)]$ is a call option on the MtM with zero strike. In the Gaussian HW setting, approximating $\text{MtM}_t \sim \mathcal{N}(m_t, s_t^2)$ with $s_t \propto \sigma$,

$$\text{EPE}(t) = m_t \, \Phi\left(\tfrac{m_t}{s_t}\right) + s_t \, \varphi\left(\tfrac{m_t}{s_t}\right), \qquad \frac{\partial \, \text{EPE}}{\partial m_t} = \Phi\left(\tfrac{m_t}{s_t}\right) > 0, \qquad \frac{\partial \, \text{EPE}}{\partial s_t} = \varphi\left(\tfrac{m_t}{s_t}\right) > 0.$$

This closed-form approximation gives approximately +1,700 EUR for the IR Delta and +2,700 EUR for the Vega, consistent with the Monte Carlo values reported above.

CR Delta. A higher hazard rate λ increases the marginal default probability $e^{-\lambda t_{i-1}} - e^{-\lambda t_i}$ on the relevant horizon, inflating the CVA. Since for small λ the CVA is almost linear in λ, a first-order check is $\Delta\text{CVA} \approx \text{CVA} \cdot \Delta\lambda\,(1/\lambda - \bar{t})$, where $\bar{t} \approx 4.2$ years is the exposure-weighted average default time. Using the sensitivity-engine base CVA of 207,600 EUR, this gives ≈ 950 EUR, in line with the Monte Carlo value of +952 EUR.

IR Delta. The +1 bp parallel shift is applied directly to the ACT/365 zero rates of the bootstrapped curve, $B'(t_0, T) = B(t_0, T)\, e^{-\Delta r\, T}$, while the fixed rate K is kept frozen at its contractual inception value. Two channels are at work:

- **Exposure channel (positive).** Every $A(s, \tau)$ is scaled by $e^{-\Delta r(\tau - s)}$, so the price of the residual coupon bond $P_s = \sum_j c_j B(s, T_j) + B(s, T_N)$ decreases in every scenario and the payer $\text{MtM}_s = N(1 - P_s)$ increases path by path (no simulated path shows a decrease; at 4 years the average increase is about 52,600 EUR). Polimi Bank pays fixed and receives floating: when rates rise, the swap is worth more to the bank, the counterparty owes more, and more is lost upon its default.
- **Discounting channel (small negative).** The higher curve lowers $B(0, t_i)$ in the CVA formula, reducing the present value of future losses.

The exposure channel dominates, yielding a net IR Delta of +1,673 EUR per bp.

Vega. The Hull–White volatility σ controls the dispersion of x_t and, through $B(s, \tau) = A(s, \tau) \exp(-x_s C(s, \tau))$, that of the simulated MtM. To avoid Monte Carlo noise of order $\mathcal{O}(1/\sqrt{N})$, comparable to the signal for a 1 bp bump on $\sigma = 94$ bp, we use common random numbers: with the same seed and $x_0 = 0$ the bumped paths are exactly the base paths rescaled by σ'/σ. The CVA is economically a short call on the MtM: a wider MtM distribution fattens the right tail, where the bank loses, while the left tail costs nothing because exposure is floored at zero. If $m_t \approx 0$, then $\text{EPE}(t) \approx s_t/\sqrt{2\pi} \propto \sigma$, so $\Delta\text{CVA} \approx \text{CVA} \cdot \Delta\sigma/\sigma$. Using the main-loop base CVA of 251,722 EUR (which is the economically correct reference for the scaling), this gives $\approx$ 2,680 EUR. A further positive contribution comes from the convexity term in $A(s, \tau)$, which decreases with σ and therefore raises the expected payer MtM. The Monte Carlo Vega is +2,731 EUR per bp, i.e. a +1.08% change in CVA for a +1.06% relative change in σ.

Errors found and corrected in the sensitivity code. A first implementation returned −32,250 EUR (IR Delta) and −41,392 EUR (Vega), with signs inconsistent with the theory above. Two errors were identified:

(i) In the original code the base CVA was taken from the main Monte Carlo loop (251,722 EUR) while the bumped CVAs were computed with `compute_epe_base`. The latter contains a residual bias: it discounts the cash flows from index `[i:]` instead of `[i+1:]`, so on the annual payment dates the fixed coupon paid at t_i (with $B(t_i, t_i) = 1$) is counted as still due, lowering the MtM by $N \cdot c \approx$ 4.43 M EUR in every scenario. This bias reduces the base CVA computed by `compute_epe_base` from the main-loop value of 251,722 EUR to 207,600 EUR (bias = −44,122 EUR). Because base and bumped were computed with different engines, the reported deltas were dominated by this bias instead of reflecting the true bump effect.

(ii) The IR bump rebuilt the discount factors from ACT/365 zero rates multiplied by ACT/360 tenors, starting from a zero-rate series already re-indexed in the MtM loop (values shifted with respect to the discount-factor dates). The effective shift was about 6–7 bp instead of 1 bp; the bump is now applied directly to the discount factors with ACT/365 tenors.

The corrected implementation uses `compute_epe_base` for both base and bumped CVAs, with the correct ACT/365 bump. The residual −44,122 EUR bias of point (i) is identical for base and bumped computations, so it cancels in the deltas: the reported sensitivities are now +952, +1,673 and

+2,731 EUR per bp, positive and of the same order of magnitude as the closed-form estimates. The base CVA used in Table 2 is the `compute_epe_base` value of 207,600 EUR, while the scenario CVAs of Table 1 remain the main-loop values (251,722 EUR for the base case), unaffected by the bump-and-reprice exercise.